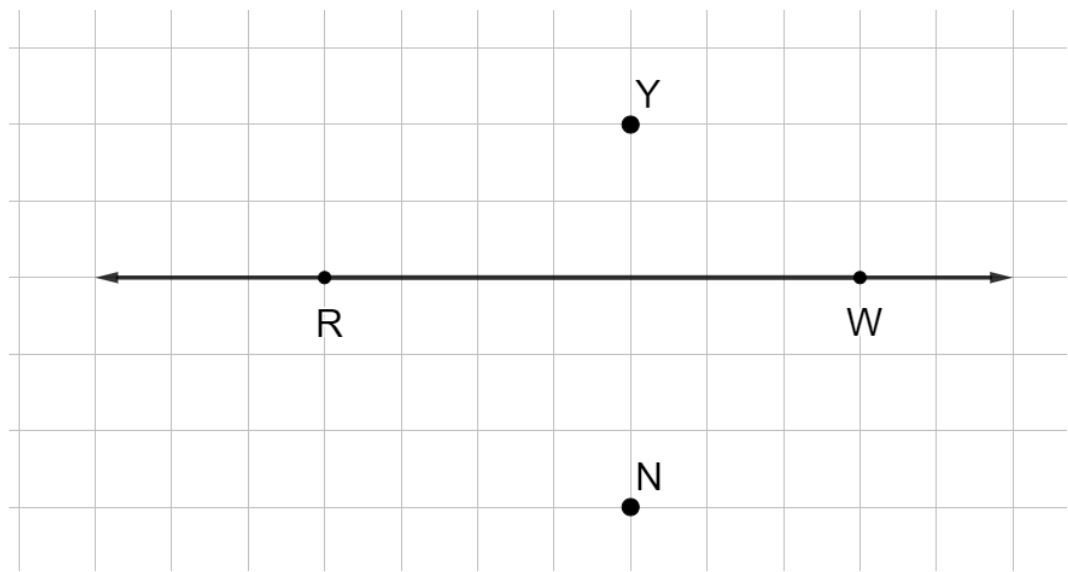


Line RW and points N and Y are shown.



1. Fold the paper so that the fold goes through points N and Y and line RW coincides with itself. Draw a dashed line on the grid that shows the fold.
2. Two students wrote a description of the relationship the dashed line has with line RW .

Olivia	Makayla
The dashed line divides line RW in half.	The dashed line makes a right angle with line RW .

Write a brief explanation that describes which student’s description is not correct.

3. Fold the dashed line on itself so that point N coincides with line RW . Draw a line along the fold and name the line BK .

Complete the statements about the construction of the dashed line and line BK using words from the word bank.

Word Bank

line

line segment

parallel

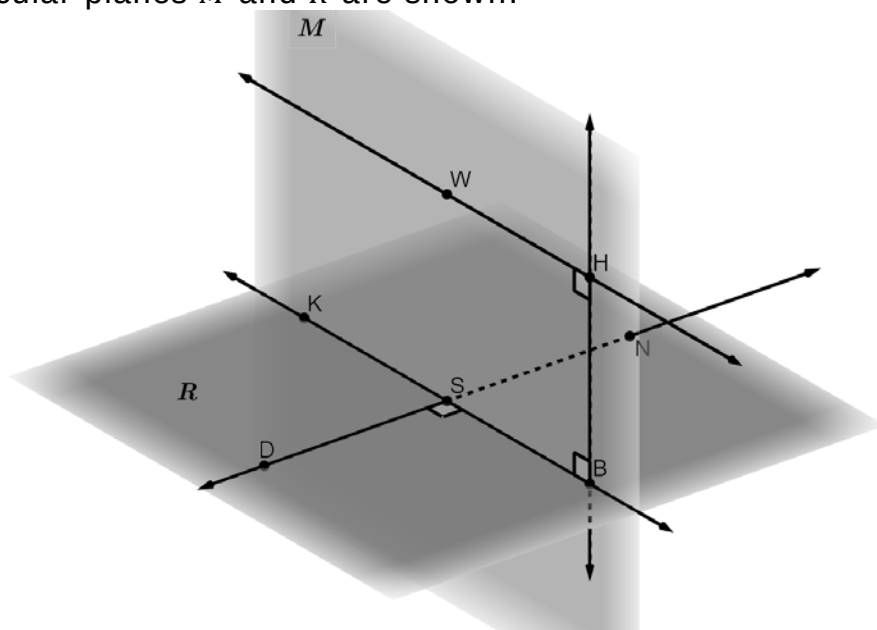
perpendicular

plane

point

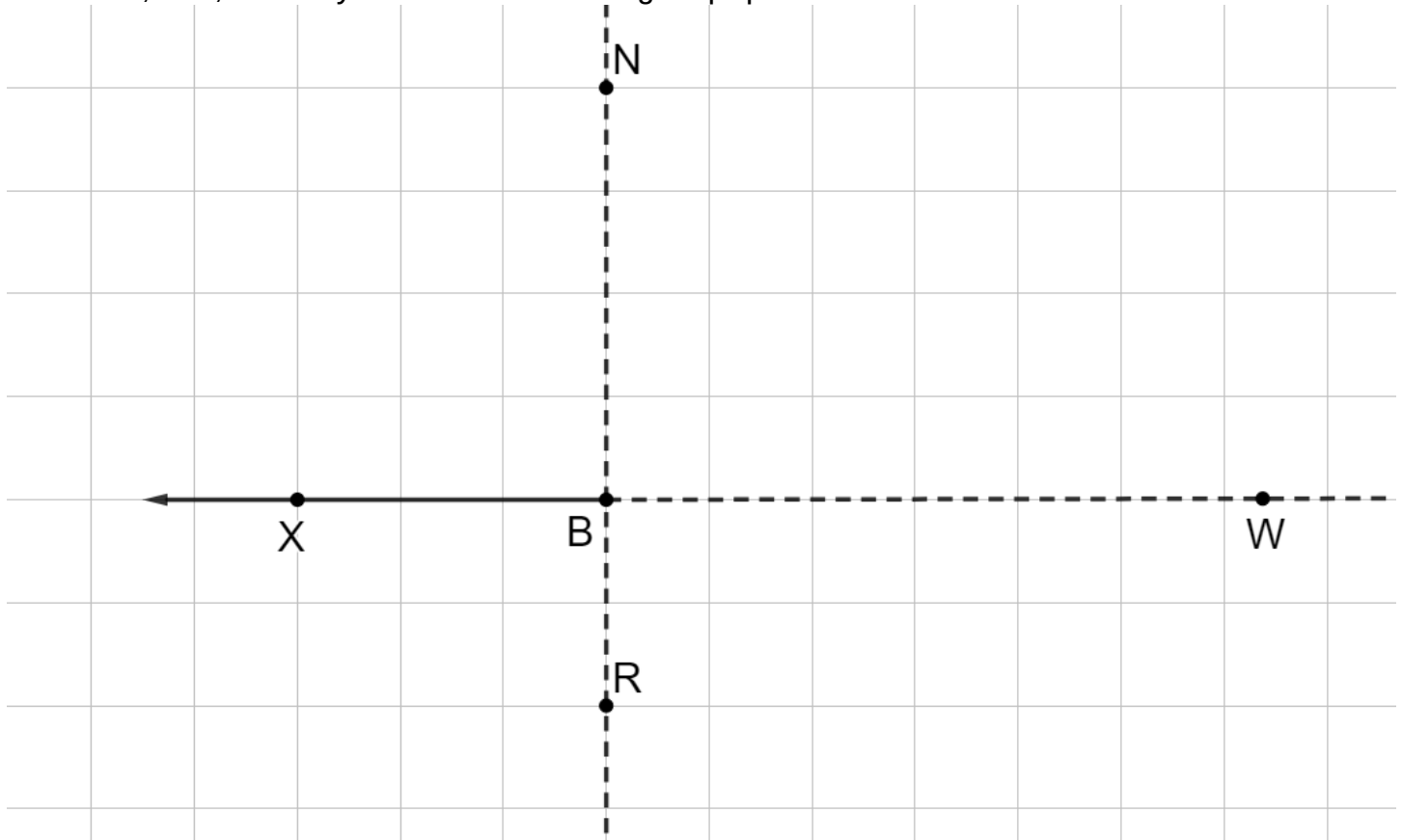
4. The homework paper represents a _____.
5. Lines RW and BK are _____.
6. The dashed line that passes through points N and Y is _____ to lines RW and BK .

Perpendicular planes M and R are shown.



7. Identify two lines that are parallel.
8. Identify two lines that are perpendicular and lay in plane R .
9. Identify two lines that are perpendicular and lay in plane M .
10. Identify two lines that are skew.

Lines RN , BW , and ray BX are shown on grid paper.



1. Suppose you have a piece of patty paper that has ray BK on it. Describe how ray BK can be used to create angle XBK such that angle XBK is obtuse.
2. Draw angle XBK on the grid paper.
3. Suppose you have a piece of patty paper that has ray BH on it. Describe how ray BH can be used to create angle XBH such that angle XBH is a straight angle.
4. Draw angle XBH on the grid paper.
5. Suppose you have a piece of patty paper that has ray BL on it. Describe how ray BL can be used to create angle XBL such that angle XBL is an acute angle.

6. Draw angle XBL on the grid paper.

Micah listed key characteristics of points, lines, and planes. His list is shown.

Point	Line	Plane
has no dimension	has infinite length with no width or height	has three dimensions
indicates a position or location	has a measure of 180°	has infinite length and width, but no height
goes on forever	goes on forever in both directions	has a rectangular shape when drawn
can vary in size	can be named using two capital letters or one lowercase letter	goes on forever

Micah's list of characteristics has a few mistakes.

7. Which, if any, of Micah's characteristics of a point is not correct?

8. Which, if any, of Micah's characteristics of a line is not correct?

9. Which, if any, of Micah's characteristics of a plane is not correct?

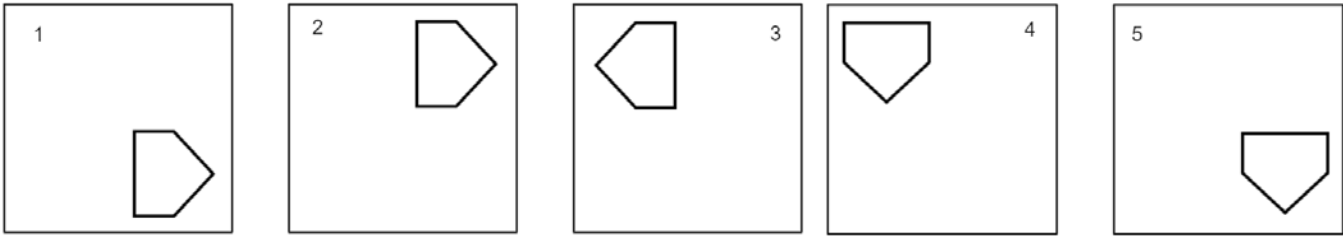
10. Select all of the terms from the word bank that are considered undefined terms in Geometry.

Word Bank					
line	line segment	parallel	perpendicular	plane	point

Geometry
Unit 1
Lesson 3 Homework

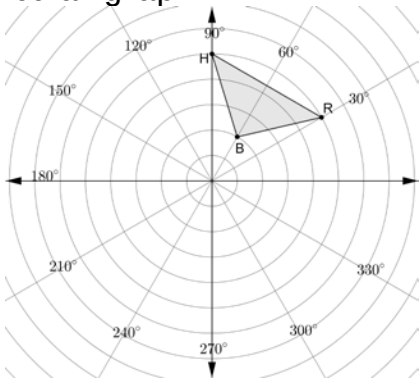
Name _____
Date _____
Period _____

Five frames that feature a polygon in different positions are shown.



	Movement from...	Description
1.	Frame 1 to Frame 2	
2.	Frame 2 to Frame 3	
3.	Frame 3 to Frame 4	
4.	Frame 4 to Frame 5	

Triangle RHB is shown on the circular graph.



5. Plot each of the points.

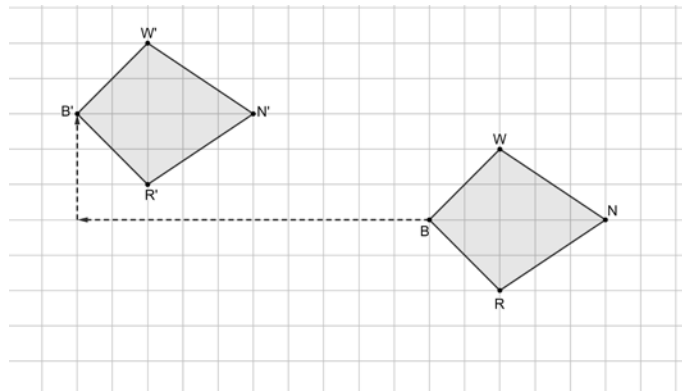
Point	Description
W	Plotted on the 120° line and on the same circle as B .
Y	Plotted on the 150° line and on the same circle as R .
K	Plotted on the 300° line and on the same circle as H .
T	Plotted on the 330° line and on the same circle as B .
G	Plotted on the 360° line and on the same circle as R .

6. Complete the statements about the triangles.

$\triangle HYW$ is a ☐ reflection
☐ rotation of $\triangle HRB$.

$\triangle GKT$ is a ☐ reflection
☐ rotation of $\triangle HRB$.

A diagram of $WNRB$ and $W'N'R'B'$ is shown. Arrows are drawn from vertex B to vertex B' .



Complete the statements.

7. The arrows show a possible path of movement. This path is called a

- ☐ reflection.
☐ rotation.
☐ translation.

8. The same path ☐ exists
☐ does not exist for all points on a diagram.

9. The path is called ☐ isometry.
☐ a mapping.

10. The transformation shown on the grid can be referred to as

- ☐ an isometry
☐ a mapping
☐ a rigid motion

or as

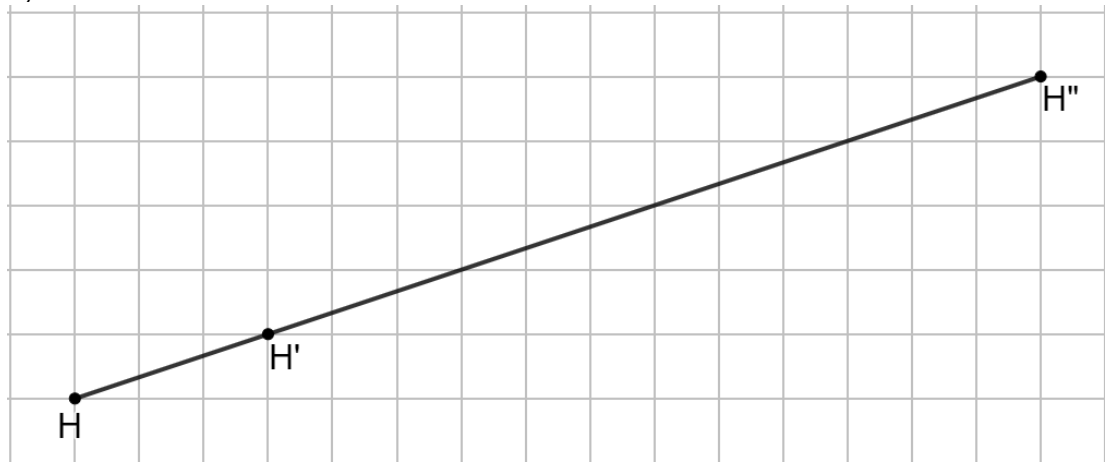
- ☐ an isometry
☐ a mapping
☐ a rigid motion

because

the transformation

- ☐ preserved distance.
☐ preserved angle measure.
☐ preserved angle measure and distance.

Points H , H' , and H'' are shown.



- 1. Determine the number of copies of $\overline{HH'}$ that will fit on $\overline{HH''}$.
- 2. Use the definition of dilation to explain why $\overline{HH''}$ is a dilation of $\overline{HH'}$.
- 3. Complete the statement.

Point

H

H'

H''

 partitions

$\overline{HH'}$

$\overline{H'H''}$

$\overline{HH''}$

 so that the ratio of $\frac{HH'}{HH''}$ is

$\frac{1}{4}$

$\frac{1}{5}$

Use the Pythagorean theorem to complete the table.

	Segment	Length
4.	$\overline{HH'}$	
5.	$\overline{H'H''}$	
6.	$\overline{HH''}$	

7. Complete the statements.

Point H' is

$\frac{1}{4}$

$\frac{1}{5}$

 of the distance from H to H'' . The scale factor that can be

used to multiply

$\overline{HH'}$

$\overline{H'H''}$

$\overline{HH''}$

 by to produce

$\overline{HH'}$

$\overline{H'H''}$

$\overline{HH''}$

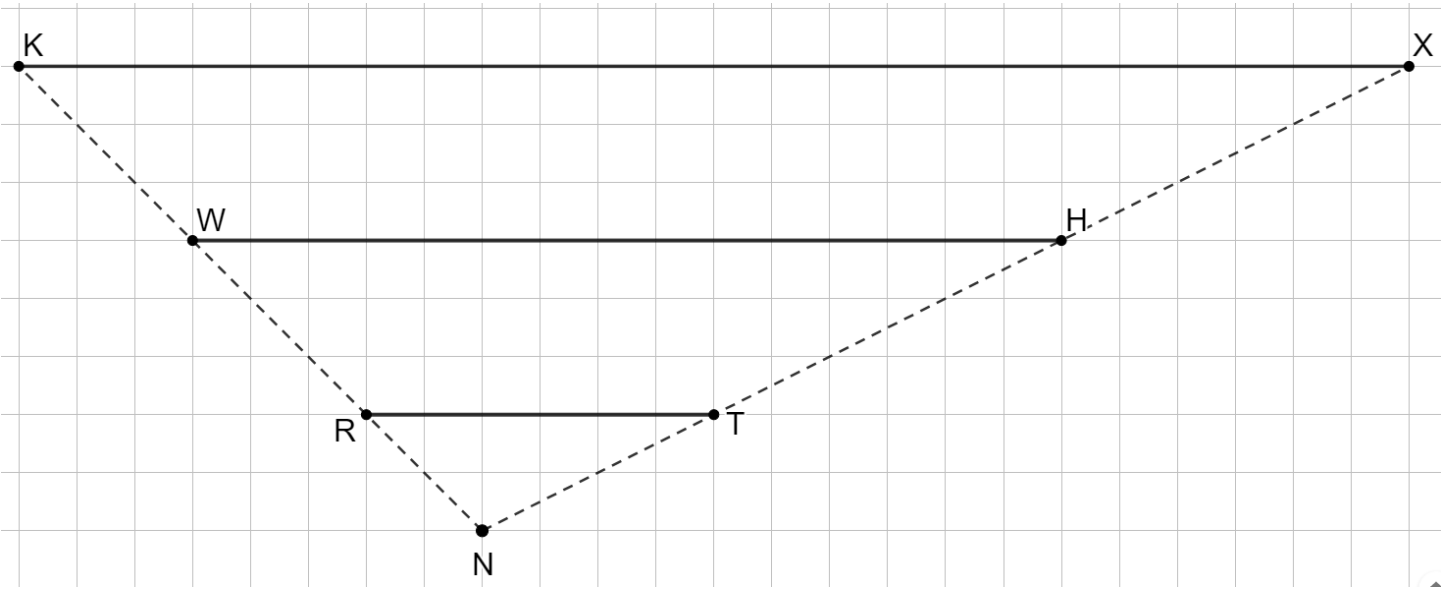
 is

$\frac{1}{4}$

$\frac{1}{5}$

.

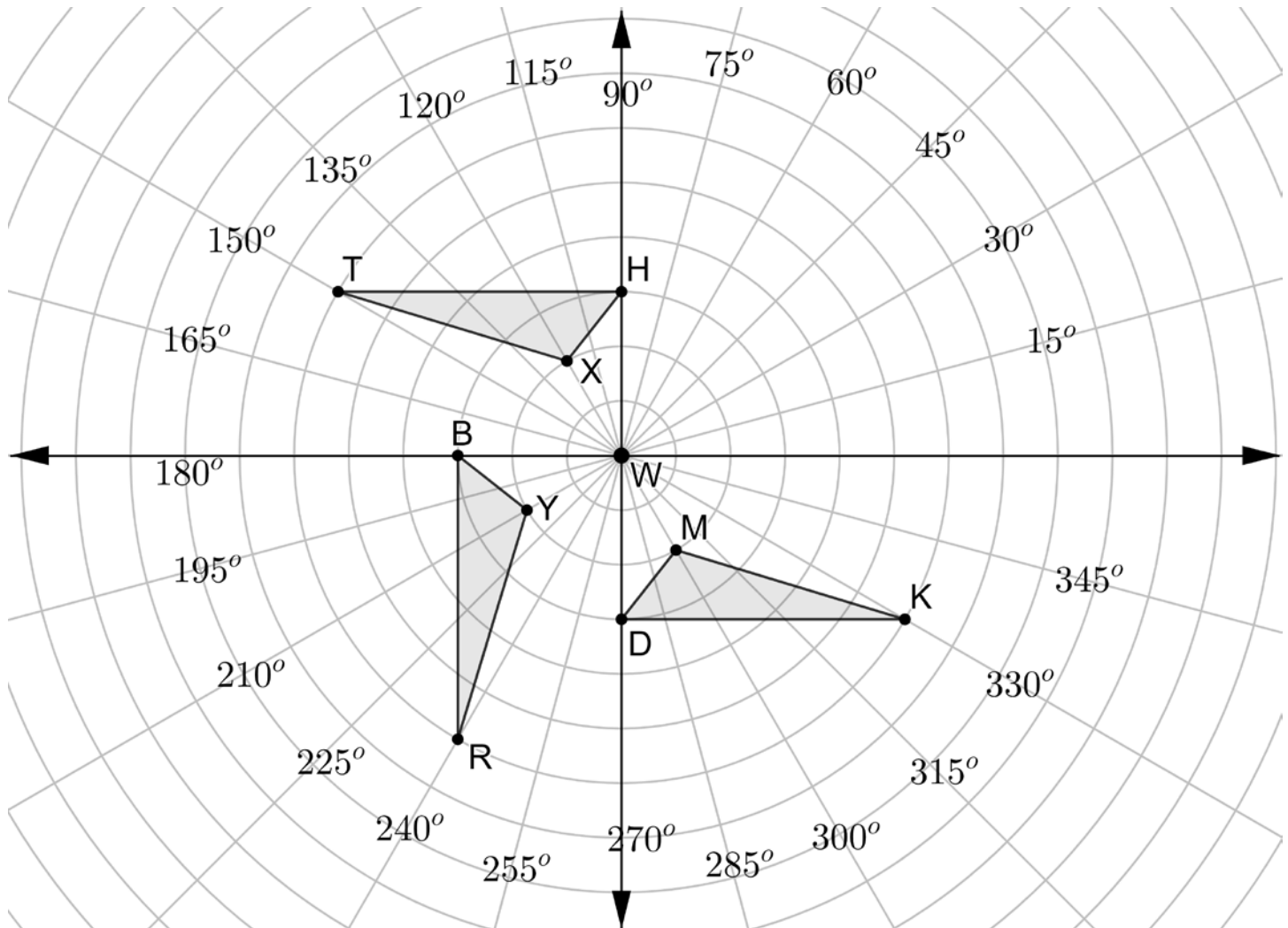
Line segments KX , WH , and RT are shown where $\overline{RT}$ is dilated using point N to create $\overline{WH}$ and $\overline{KX}$.



Determine each of the ratios.

	Ratio	Length
8.	$NR:NW$	
9.	$RT:WH$	
10.	$RT:KX$	

Triangles THX , YBR , and DKM are shown on the circular grid.



Draw each of the angles on the grid.

1. $\angle XWY$

4. $\angle BWD$

2. $\angle KWR$

5. $\angle BWH$

3. $\angle DWH$

6. $\angle MWX$

7. Select all of the angles drawn on the grid whose measure is 90° .

- ☐ $\angle XWY$
- ☐ $\angle KWR$
- ☐ $\angle DWH$
- ☐ $\angle BWD$
- ☐ $\angle BWH$
- ☐ $\angle MWX$

8. Select all of the angles drawn on the grid whose measure is 180° .

- ☐ $\angle XWY$
- ☐ $\angle KWR$
- ☐ $\angle DWH$
- ☐ $\angle BWD$
- ☐ $\angle BWH$
- ☐ $\angle MWX$

Complete the statements.

9. $\triangle THX$ is created by rotating

- ☐ $\triangle KDM$
- ☐ $\triangle RBY$

 180° clockwise about point W .

10. $\triangle RBY$ is created by rotating

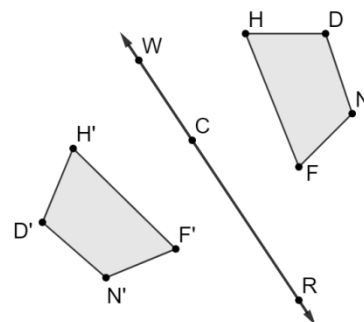
- ☐ $\triangle KDM$
- ☐ $\triangle THX$

 90° clockwise about point W .

Quadrilaterals $HDNF$ and $H'D'N'F'$ are shown.

1. Complete the table to describe the single transformation that was used to create $H'D'N'F'$.

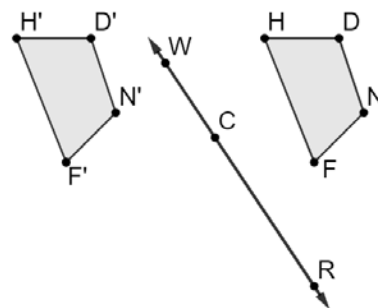
Type	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	



Quadrilaterals $HDNF$ and $H'D'N'F'$ are shown.

2. Complete the table to describe the single transformation that was used to create $H'D'N'F'$.

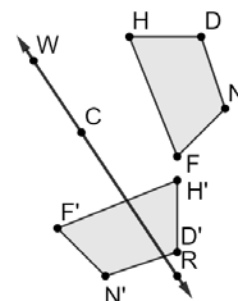
Type	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	



Quadrilaterals $HDNF$ and $H'D'N'F'$ are shown.

3. Complete the table to describe the single transformation that was used to create $H'D'N'F'$.

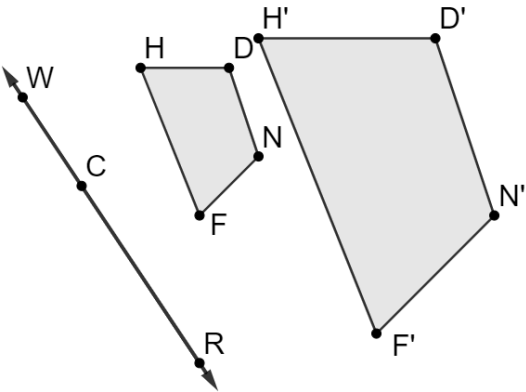
Type	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	



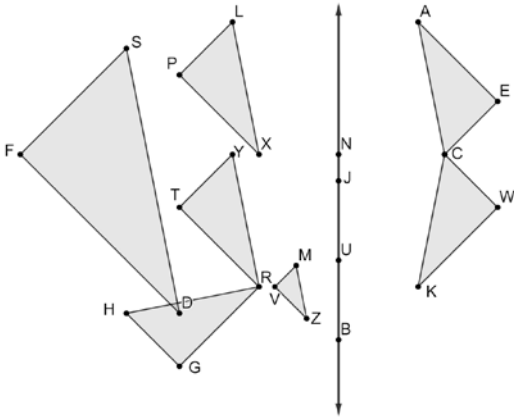
Quadrilaterals $HDNF$ and $H'D'N'F'$ are shown.

4. Complete the table to describe the single transformation that was used to create $H'D'N'F'$.

Type	Center, Line, or Movement	Degrees or Scale Factor
Dilation Rotation Reflection Translation	Center of dilation is _____. Center of rotation is _____. Line of reflection is _____. Down Left Right Up	



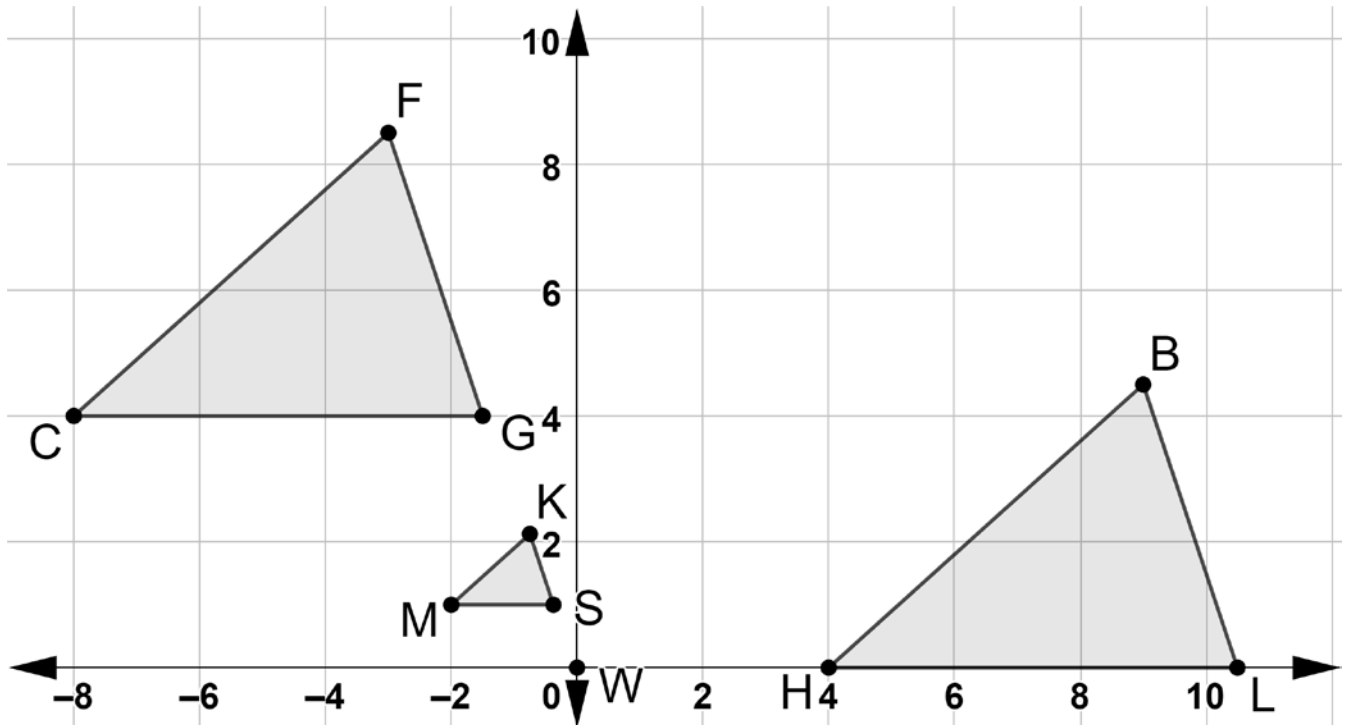
The diagram shows six different single transformations of $\triangle YTR$.



Complete the table.

Triangle	Type of Transformation
5. $\triangle MVZ$	
6. $\triangle HGR$	
7. $\triangle CWK$	
8. $\triangle LPX$	
9. $\triangle SFD$	
10. $\triangle CEA$	

Three triangles are shown on the coordinate grid. $\triangle LBH$ and $\triangle SKM$ are created using a single transformation of $\triangle GFC$.



Complete each statement using an angle or a segment from $\triangle GFC$.

1. $\overline{LB} \cong$ _____

2. $\overline{SM} \sim$ _____

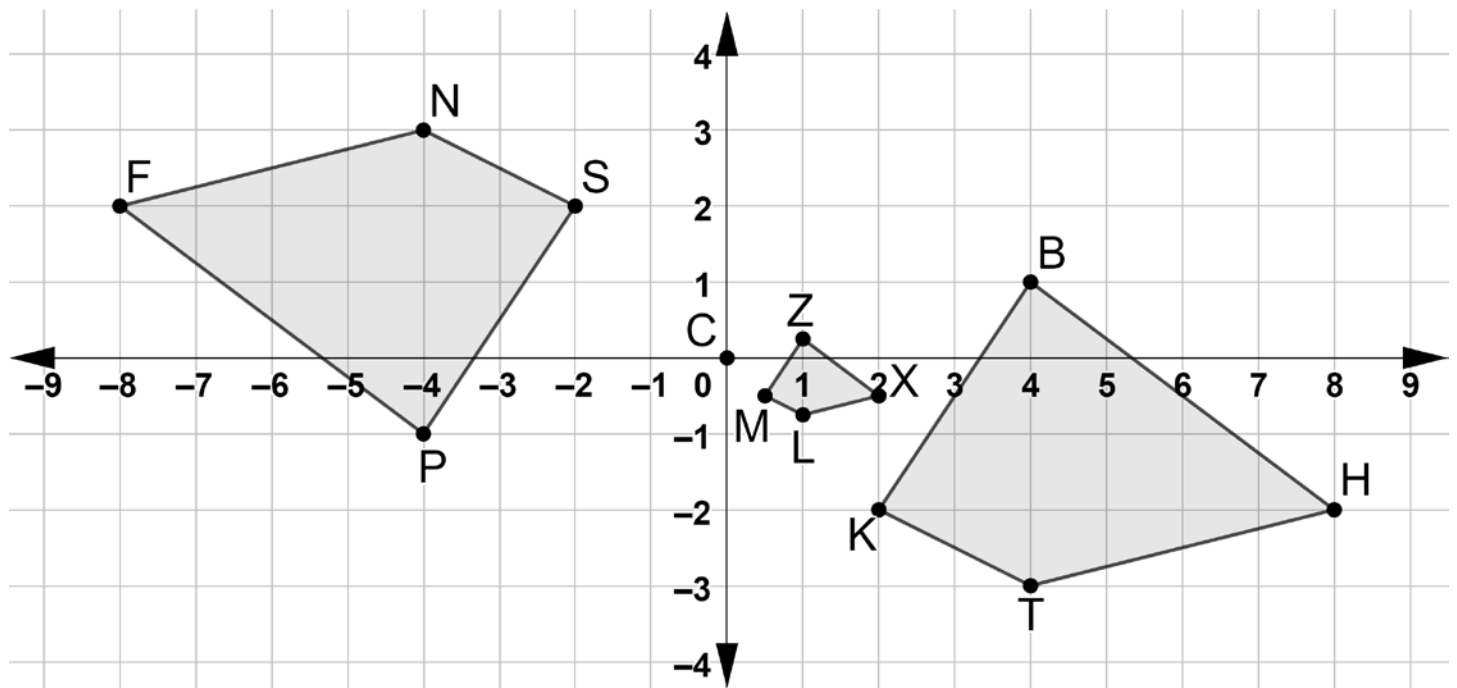
3. $\overline{BH} \cong$ _____

4. $\overline{LH} \cong$ _____

5. $\overline{MK} \sim$ _____

6. $\overline{SK} \sim$ _____

Three quadrilaterals are shown on the coordinate grid. Quadrilaterals $PFNS$ and $ZXLM$ are created using a single transformation of quadrilateral $BHTK$.



Complete the following statements with the $\cong$ or $\sim$ symbol.

7. $\angle MLX$ _____ $\angle KTH$

8. $\overline{HB}$ _____ $\overline{PF}$

9. $\overline{ZM}$ _____ $\overline{BK}$

10. $\angle BHT$ _____ $\angle NFP$

Lesson 8 Homework

Period _____

A horizontal number line with four points marked by dots. Above the dots are the labels W, X, Y, and Z from left to right. The points are evenly spaced, representing a scale from 0 to 3.

1. Find the value of x .
2. Find the length of $\overline{WX}$.
3. Find the length of $\overline{XY}$.
4. Find the length of $\overline{XZ}$.

P , Q , and R are three collinear points that exist such that $\overline{PQ}$ and $\overline{QR}$ are congruent.
 $PQ = 6x - 8$ and $QR = 10x - 44$.

5. Draw a diagram that represents points P , Q , and R .

6. Find the value of x .

7. How long is segment $\overline{PQ}$?

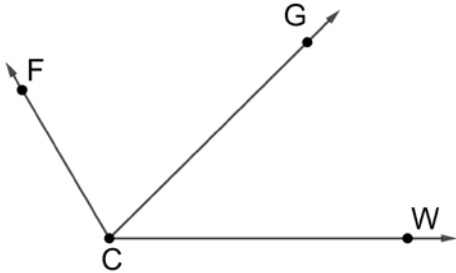
8. How long is segment $\overline{QR}$?

9. How long is segment $\overline{PR}$?

10. Use a compass and straightedge to make a copy of this segment.



Angle FCW is shown. $m\angle FCW = 120^\circ$, $m\angle FCG = (3x + 4)^\circ$ and $m\angle GCW = (2x - 1)^\circ$.

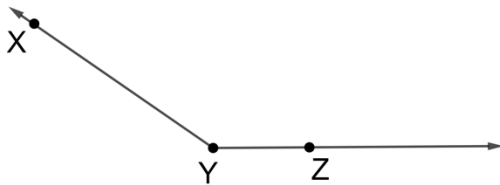


1. Find the value of x .

2. Find $m\angle FCG$.

3. Find $m\angle GCW$.

4. Use a straightedge and a compass to copy $\angle XYZ$.



Given that ray MS lies inside $\angle RMT$ such that $m\angle RMT = (10x)^\circ$, $m\angle RMS = (2x - 5)^\circ$ and $m\angle SMT = (-2x + 145)^\circ$.

5. Draw a diagram that represents $\angle RMT$ and $\overrightarrow{MS}$.
6. Find the value of x .
7. Find $m\angle RMT$.
8. Find $m\angle RMS$.
9. Find $m\angle SMT$.
10. $\overrightarrow{DN}$ lies inside $\angle LDP$. If $\angle LDP$ is a straight angle, and $m\angle LDN$ is five times $m\angle NDP$, find $m\angle LDN$.

Geometry

Name_____

Unit 1

Date_____

Lesson 10 Homework

Period_____

$\overline{LM}$ bisects $\overline{ZT}$ at point S. $\overline{ZS}$ has a length of $3x + 7$, and $\overline{ST}$ has a length of $-5x + 23$.

1. Draw a diagram of the description.

2. Find the value of x .

3. Determine the length of $\overline{ZT}$.

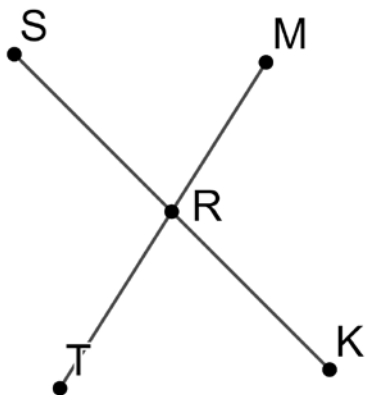
$\overline{XY}$ bisects $\overline{WB}$ at point L . $WB = 3x + 8$ and $WL = \frac{1}{2}x + 12$.

4. Draw a diagram of the description.

5. Find the value of x .

6. Find the length of $\overline{WB}$.

$\overline{MT}$ bisects $\overline{SK}$ at point R as shown. $\overline{SK}$ has a length of $24x - 12$, $\overline{RK}$ has a length of $10x - 2$, $\overline{TR}$ has a length of $4x + 5$, and $\overline{RM}$ has a length of $8x - 5$.

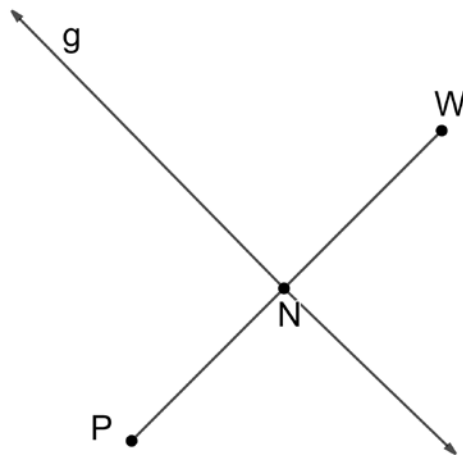


7. Based on the given information which two segments are congruent?
8. Find the value of x .
9. Find the length of $\overline{RK}$.
10. Find the length of $\overline{MT}$.

1. Use a compass and a straightedge to construct a perpendicular bisector of $\overline{XY}$.

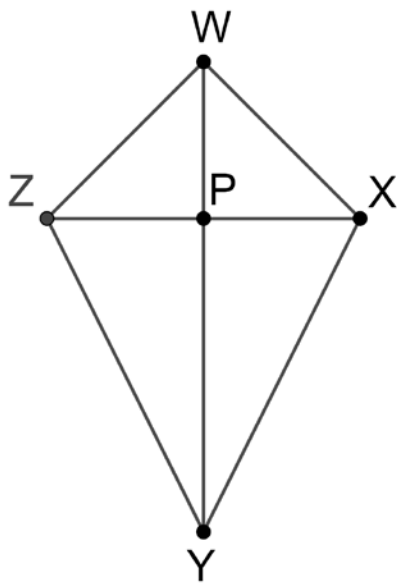


Line g is a perpendicular bisector of $\overline{PW}$ at point N .



2. If $PN = 4x + 17$ and $WN = 9x - 20$, find the value of x .
3. How long is $\overline{PN}$?
4. How long is $\overline{PW}$?

A quadrilateral is shown where $\overline{WY}$ is a perpendicular bisector of $\overline{ZX}$.



5. What angle is congruent to $\angle WZP$?
6. What angle is congruent to $\angle ZWP$?
7. Which angle(s) are congruent to $\angle WPZ$?
8. What kind of triangle is $\triangle XYZ$?

$\overline{SL}$ is a perpendicular bisector of $\overline{RT}$, intersecting at point N . The length of $\overline{SR}$ is $\frac{2}{3}x - 40$ and the length of $\overline{ST}$ is $\frac{1}{4}x + 12$.

9. Find the value of x .
10. Find the length of $\overline{ST}$.

Geometry

Unit 1

Lesson 12 Homework

Name_____

Date_____

Period_____

$\angle MPS$ is bisected by $\overrightarrow{PN}$. $m\angle MPN = (7x + 4)^\circ$ and $m\angle NPS = (16x - 68)^\circ$.

1. Draw a sketch that represents the description.

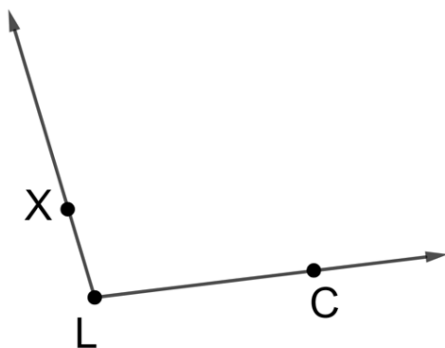
2. Find the value of x .

3. Determine $m\angle MPN$.

4. Find $m\angle MPS$.

5. $\angle FBW$ is bisected by $\overrightarrow{BR}$ where the $m\angle FBW = (20x - 14)^\circ$ and $m\angle RBW = (\frac{5}{2}x + 8)^\circ$. Find $m\angle FBR$.

6. Use a compass and a straightedge to bisect $\angle XLC$.



$\angle DMF$ is bisected by $\overrightarrow{MG}$ where $m\angle DMG = (4x)^\circ$ and $m\angle DMF = (2x + 24)^\circ$.

7. Sketch a diagram that could be used to represent this situation.

8. Find the value of x .

9. Find $m\angle DMG$.

10. Find $m\angle DMF$.